NeuroWorks

AI Digital Workforce and Intelligent Organisation Platform

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Team Name: NeuroWorks
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> **FORMAT COMPLIANCE**
> Body: 5 required sections, Arial 11pt, 1.15 line spacing, 1-inch margins, ≤10 pages excluding this cover and the appendices. Submit as PDF only. File named Neuroworks_AI4I_Proposal_Development.pdf.

Section 1 — Problem Definition & Strategic Alignment

1.1 The Problem

Zimbabwean SMEs, NGOs, and government-adjacent institutions face a structural capacity gap, not just a tooling gap: the volume of knowledge work — reconciliation, reporting, compliance, coordination — structurally exceeds what an affordable number of hires can cover. A single NGO programme officer routinely performs the work of several specialists simultaneously; quality suffers, deadlines slip, compliance fails. Hiring is slow and expensive relative to the pace institutional mandates require. Job descriptions, qualifications, and org charts are industrial-era artefacts poorly suited to closing this gap.

1.2 Target Users and Beneficiaries

NeuroWorks is an operational B2B/B2G platform, not a consumer product. Twelve built-in sector context packs name real Zimbabwe-market institutional targets:

Sector	Named target users / institutions
Finance / fintech	Reconciliation desks, agent-network managers — RBZ framework, EcoCash, OneMoney
Agriculture	Field-reporting teams, input-financing coordinators — Pfumvudza/Intwasa, GMB
Health	Community health worker coordinators — MOHCC, DHIS2-integrated facilities
Education / Public services	Admin and reporting staff — ZIMSEC, PRAZ, ZIMRA, devolution departments
Tourism / Informal economy	Booking desks — ZTA, UNIVISA/KAZA; Mbare Musika market operators
Development / NGO	Grant reporting, donor coordination — WFP, WHO, UNICEF, UNDP cadences

1.3 Product Function and Expected Value

NeuroWorks turns a job description into a digital worker: an LLM-powered agent that receives a goal, forms a plan, calls tools through a governed allowlist, and pauses for human approval before anything consequential executes. It is briefed like a person and supervised like a new hire, not configured like software. Expected value is measured directly, not asserted: task completion rate, time saved against the prior manual process, cost per task, and non-technical operator adoption (§3.4, §5).

1.4 Strategic Alignment

Zimbabwe National AI Strategy

- **Talent and Capacity Development**: NeuroWorks turns job descriptions into AI workers that perform roles at scale, and IntelliNexus turns agent output into governed, hash-verified institutional data — a continuous capacity-building loop for the humans who supervise and learn alongside digital workers.
- **Computational and Data Sovereignty**: the default architecture runs inference locally on operator-owned hardware (62.7% of measured calls served on a consumer GPU, §2.5); no data leaves the machine unless an operator explicitly opts into cloud escalation.
- **AI Adoption and Service Transformation**: deployed AI workforce under an institution's own uploaded policies, not a pilot chatbot — pre-execution governance, mandatory human-in-the-loop approval, full audit trail, named institutional sector packs (PRAZ, MOHCC, ZTA).

— **Governance and Ethics**: pre-execution policy enforcement, mandatory HITL, full audit trail, and AES-256-GCM credential encryption are the mechanics of governance, not a claim bolted onto a capability demo (§4).

NDS2 (2026–2030) — verified against the primary document

NDS2 organises its programme around ten national priority areas (down from fourteen under the prior strategy), one per chapter (Ch. 3–12), each led by a Thematic Working Group. NDS2 places AI adoption directly under Priority 5 — Science, Technology, Digital, Innovation and Human Capital Development — committing that AI adoption "will be guided by the National AI Strategy" and naming "strengthening AI governance frameworks across all sectors of the economy" as an explicit intervention (NDS2 ¶1152–1153). NDS2 separately commits to expanding the Zimbabwe Centre for High-Performance Computing for, among other workloads, artificial intelligence (NDS2 ¶1101–1104) — the same CCE pathway addressed in §3.3. NeuroWorks maps to five of the ten priority areas:

NDS2 priority area (chapter)	Fit	Evidence
Ch.7, Priority 5 — Science, Technology, Digital, Innovation & Human Capital Development	Direct	Operationalises NDS2's named AI-governance commitment (¶1153) via pre-execution policy enforcement and full audit trail; ZCHPC/CCE pathway matches NDS2's own HPC-for-AI expansion plan (¶1101–1104)
Ch.12, Priority 10 — Governance, Institution Building, Peace & Security	Direct	Governance engine, mandatory HITL, full audit trail on every agent action
Ch.9, Priority 7 — Regional Development & Inclusivity through Devolution & Decentralisation	Direct	Named alignment to PRAZ and devolution-agenda departments (§1.2)
Ch.3, Priority 1 — Macro-Economic Stability & Financial Sector Deepening	Partial	Reconciliation workflows for fintech desks; NDS2 names fintech/digitalisation expansion under Financial Sector Deepening (¶286–290)
Ch.10, Priority 8 — Social Development, Gender & Social Protection	Partial	Community-health-worker coordination in the health sector pack (§1.2)

No claim is made against the remaining five priority areas (Inclusive Economic Growth & Structural Transformation; Infrastructural Development & Housing; Agriculture, Food, Climate & Environment; Job Creation, Youth Entrepreneurship, Creative Industry, Sport & Culture; Image Building, International Relations & Trade). Stated plainly rather than stretched to fit.

Section 2 — Technical Design & Product Logic

2.1 System Architecture

Three layers (full diagram: Appendix A). Layer 1 — tools and infrastructure (PostgreSQL, Qdrant, Ollama, Redis, payment/comms rails). Layer 2 — HERMES, an MCP tool-governance layer of ~90 allowlisted primitives across 12 namespaces defining exactly what agents can and cannot do. Layer 3 — LangGraph multi-agent digital workers that plan, call tools through HERMES, and stop at a mandatory HITL gate before consequential actions execute.

Component	Detail
Backend	Node.js / TypeScript, Express — 28 live routers, 85 lib files, 46 route handlers. No scaffolding stubs.
Frontend	Next.js 14 / React / TypeScript — 37 pages (console, Mission Control, governance manager, cost dashboard).
Orchestration	LangGraph multi-agent; structured audit logging on every action.
Inference	Ollama (local, default) + OpenRouter/Claude (cloud, opt-in, circuit-breaker fallback to Ollama).
Data	PostgreSQL (relational), Qdrant (RAG, org-isolated, HNSW), Redis (working memory/queues).
Secrets	AES-256-GCM encrypted store (secret-box.ts); never logged or passed by value into agent context.

2.2 AI Justification & Model Rationale

Per capability, stated against the ToR's sledgehammer test — would ordinary rules or SQL suffice?

Capability	Why AI, not rules/SQL	Model tier used
Reconciliation synthesis across 24 DB tables vs. a natural-language policy doc	Requires interpreting an unstructured policy document alongside structured data — not expressible as a fixed query	Local 7B default; escalates only on quality.check failure
Donor-report narrative drafting from an uploaded PDF	Free-text synthesis with institution-specific compliance framing — no fixed template covers this	Local 7B default; cloud opt-in for long/complex documents
Shona/Ndebele service-policy chat	Natural-language understanding across two under-resourced languages within dynamic policy bounds	Local, glossary-grounded — no external API
Policy-document → structured constraint extraction	Converts arbitrary prose policy into hard/soft rule sets — a genuinely open-ended language task	Local LLM, human-reviewed before activation
Routine data lookups (org.lookup, db.list_tables)	Deliberately NOT routed through an LLM — plain function calls	No model — direct primitive, by design

The last row is included deliberately: NeuroWorks does not route every action through an LLM. Primitives with a deterministic answer are implemented as plain function calls; the model is invoked only where language understanding, multi-step planning, or context synthesis is genuinely required. This is the concrete evidence against forced or cosmetic AI use.

2.3 Dataset Statement & Provenance

Source	Type	Size / status	Licence	Limitations
Production PostgreSQL	Real, operator-owned	1 org connected live, 11 Jul 2026	Operator data; NeuroWorks is processor, read-only by default	Single connected source to date
Awesome curated-software dataset	Real, third-party, ingested via IntelliNexus	682 records	CC0 — verified against source repo metadata before ingestion	Software-domain only; not Zimbabwe-sector data
Nightly self-reflection series	Real, NeuroWorks' own operational telemetry	Growing, daily	Internal	Operational metadata, not user/beneficiary data
Zimbabwe sector knowledge packs (Stream D, in build)	To be collected — public-domain and institution-provided sources	5 packs planned	Planned release: CC BY 4.0	Not yet ingested; content sourcing not yet finalised

NO SYNTHETIC DATA GENERATION

NeuroWorks does not use GAN, diffusion, or other synthetic data generation methods anywhere in the current build. Test fixtures for the 118-test unit suite use hand-constructed sample records, not generative synthesis — stated plainly to avoid any ambiguity under this criterion.

2.4 User Interaction Plan

Operators interact through the Next.js console (37 pages) or brief a worker in natural language, the way they would brief a new hire: select a sector/language pack, state a goal in plain language, upload a policy document (auto-converted to constraints), approve or decline the worker's first proposed action at the HITL gate, then review output in their own language. No prompt engineering or technical configuration is required for this path; a full non-technical-operator walkthrough is at §4.4.

2.5 Local-Inference Resource Envelope

NOT AN EMBEDDED/EDGE DEPLOYMENT

NeuroWorks' local inference runs on operator-owned GPU workstation or server hardware (Ollama), not an embedded or mobile device. The ToR's edge bar (<256MB RAM, <100ms latency) targets embedded/mobile AI and does not govern this submission's inference tier. The figures below are the correct, measured comparison for a local/on-premise GPU deployment.

Metric	Measured / specified value
Reference hardware	Consumer RTX 3060, 12GB VRAM
Default local models	Llama 3.2 3B / Qwen 2.5 7B (quantised)
Share of calls served locally (measured, 10	62.7% (640 / 1,021 calls)

days)	
Measured cloud-escalation cost	USD 8.31 / 1,021 calls (~USD 0.10/task on a fully-documented day)
Optional high-capability local tier	Llama 3.3 70B quantised, for 48GB+ VRAM hosts — no foreign data transmission

2.6 API Layer, Integrations & Governance Engine

All external integrations are exposed as named HERMES primitives; no agent can call outside this allowlist, and writes/communications require explicit HITL approval. Operators upload policy documents; a local LLM converts them into hard/soft constraint sets, displayed for review before activation and enforced pre-execution at the HITL gate — 20 categories of consequential tool calls are currently gated this way.

Section 3 — Deliverables & CCE Implementation Roadmap

3.1 Current Build Status (live, verifiable)

28 live API routers with real handlers (no scaffolding stubs) · ~90 primitives across 12 namespaces · 141 skill playbooks, no orphans · one production PostgreSQL connected live and read-only · IntelliNexus dataset published with a verifiable Merkle root hash · governance engine enforcing 20 categories pre-execution · Mailjet, Paynow Zimbabwe, and Stripe all live-tested · 118 unit tests passing · dependency manifest locked via package-lock.json at repository root · structured, incremental Git commit history (not single-commit uploads).

3.2 Delivery Streams

Stream	Weeks	Deliverable
B — Onboarding UX	1–3	Sector/language selection, personalised dashboard, Shona/Ndebele i18n strings
C — Governance UI	3–4	Policy upload, constraint review, hard/soft marking, HITL violation flagging
D — Localisation	5–6	Shona/Ndebele agent prompts; five sector packs ingested via IntelliNexus
E — Platform integrity	7–8	Cost dashboard, full audit-log replay, SkillForge code-review step
F — Polish & docs	9–12	Mobile-responsive console, governance framework doc, pilot plan

3.3 Milestones, Compute, and ZCHPC CCE Position

Milestone	Deliverable	Compute / dependencies	When
M0 — Challenge demo	Live: db.* query on production data, Mailjet send, Paynow flow, local inference, governance gate firing on a planted violation. 118 tests green.	Team hardware (RTX 3060). No ZCHPC CCE required.	31 Jul 2026
M1 — Contract signing	Stream B in staging; dependency manifest locked.	Same hardware; Qdrant/PostgreSQL/Redis local or free-tier hosted.	Week 1
M2 — First tranche	Streams C & D live; native-speaker-reviewed prompts; 5 sector packs ingested.	Managed Qdrant; OpenRouter key for quality testing.	Week 6
M3 — Pilot launch	10–20 pilot orgs onboarded; first non-technical operator case study.	Hosted PostgreSQL + Redis + Qdrant, est. USD 40–60/month.	Weeks 7–12

NeuroWorks does not require the ZCHPC CCE for primary operation — the default path runs Ollama on operator-owned hardware. For pilot organisations without GPU hardware, ZCHPC access would provide shared local inference in a controlled, data-sovereign environment. If accessed: 1× GPU node (16GB+ VRAM), 32GB RAM, 100GB storage, outbound access for

GitHub API and optional OpenRouter fallback — all dependencies declared in package-lock.json at repository root.

3.4 Pilot Evidence Plan

No signed pilot exists at submission. Rather than assert impact, this is the dated path to first evidence: by 20 July, shortlist five candidate organisations from the named sector-pack institutions (§1.2); by 24 July, secure at least one verbal agreement to run a scoped, time-boxed reconciliation task on the organisation's own data during bootcamp week; during 27 Jul–1 Aug, run it live and capture before/after time and error-rate metrics directly, not self-reported; by 1 Aug, convert to a signed LOI for the Weeks 7–12 pilot slot (§3.3, M3), or disclose specifically why it did not convert.

Section 4 — Compliance & Risk Mitigation

Written candidly: where a risk is mitigated, the mitigation is named and locatable in the code; where it is not, that is stated plainly.

4.1 Ethical Safeguards

Hallucination is mitigated at four layers: anti-fabrication rules requiring inline evidence citation; quality.check scoring every non-trivial answer with a rescue pass on low scores; peer.review sending low-confidence drafts to a second model; and a DATA GAP injection forcing disclosure when a required source was unavailable. Bias has not been independently evaluated against Zimbabwe-context prompts; a structured bias probe (gender, urban/rural, English/Shona/Ndebele framing) is scheduled before Stream D sector packs reach a pilot organisation. Shona/Ndebele output is functionally verified but not yet reviewed by a native-speaker linguistic authority — scheduled within Stream D, logged as a governance-engine gate a pilot cannot pass without.

4.2 Data Protection Act (Chapter 12:07) Compliance

DPA obligations sit with the connecting operator as data controller; NeuroWorks acts as processor on read-only instructions. Connections are read-only by default; credentials are encrypted at rest (AES-256-GCM, secret-box.ts); no data is exported, cached beyond mission scope, or published via IntelliNexus without explicit operator instruction; every query against a connected source is captured in a full audit trail. In the default configuration (local Ollama, local vault, local Qdrant) personal data never leaves the operator's machine.

4.3 Data-Use Consent

> **CURRENT STATUS**
>
> A dedicated data-use consent checkbox in the onboarding UI is not yet implemented — stated plainly rather than implied. Consent is currently enforced structurally: connections default to read-only, and no external publication of operator data occurs without an explicit, logged instruction per action. An explicit consent checkbox, presented at first data-source connection and logged with a timestamp against the org record, is scheduled for Stream B (§3.2, Weeks 1–3), ahead of the Weeks 7–12 pilot cohort.

4.4 Testing, Validation, and Non-Technical Operator Check

118 unit tests pass. The governance gate was tested against planted policy violations: it fires correctly and surfaces the specific violated rule before any action executes. A secret-pattern scanner detected a planted fake token in under 1ms. The cost tracker was audited and a 19% undercount bug found and corrected (11 Jul 2026).

Step (non-technical operator persona: rural clinic administrator)	What they see
1. Select sector + language from a dropdown	No visibility into routers, primitives, or stack
2. Brief the worker in plain language	No prompt engineering required
3. Upload existing MOHCC reporting guideline as PDF	Governance engine converts it to constraints automatically
4. Approve first proposed action (HITL gate)	Plain-language summary; one-tap

Step (non-technical operator persona: rural clinic administrator)	What they see
	approve/decline
5. Read the weekly summary in their own language	No log/hash inspection needed unless something looks wrong

4.5 Cybersecurity Protections

A real credential incident is disclosed deliberately: a GitHub PAT was inadvertently displayed in terminal output during development, caught immediately, rotated within the same session, old credential revoked. Origin-guard middleware defends against DNS rebinding and cross-origin attacks. code.exec is unsandboxed and disabled by default (explicit env flag required to enable); SkillForge-generated code runs in an E2B isolated sandbox before any operator approval or persistence. Asset and licence register: all third-party libraries, models, and APIs are declared with licence positions in the repository's Asset and Licence Register (Appendix B).

Section 5 — Sustainability & Future Adoption

5.1 Cost Projections (measured, not estimated)

Period	Total calls	Free/local	Paid cloud	Cloud cost
10 days, 3–11 Jul 2026	1,021	640 (62.7%)	381 (37.3%)	USD 8.31 total
Single day, 10 Jul 2026	~35	~22	~13	USD 3.03 (~USD 0.10/task)

5.2 Operating Cost Model and Break-Even (illustrative, to refine with real payroll)

Estimated monthly operating cost ≈ USD 21,000 (5 FTE blended team cost ~USD 17,500; hosting ~USD 50; cloud-inference buffer ~USD 150; ~15% overhead ~USD 3,300). Priced against this: Growth tier at USD 105/month breaks even at 200 organisations; Institutional tier at USD 300/month (~USD 3,600/yr annual licence, incl. support and deployment assistance) breaks even at 70 organisations — both converging on the same ~USD 21,000/month target independently, so the model does not depend on hitting both simultaneously. Every installation still runs Ollama by default at zero mandatory cloud spend; a managed relay for hard-to-self-host pieces is planned, not yet built.

5.3 Licensing Registry

NeuroWorks' code is proprietary to Aiia Rubiem and RUBIEM Developers, all rights reserved (full licence at repository root). POTRAZ is granted a non-exclusive, royalty-free right to evaluate, demonstrate, and pilot the submission for public-interest purposes; commercial use beyond the Challenge requires a separate agreement. Third-party libraries, models, and APIs are declared in the Asset and Licence Register (Appendix B). Zimbabwe sector knowledge packs will be released under CC BY 4.0.

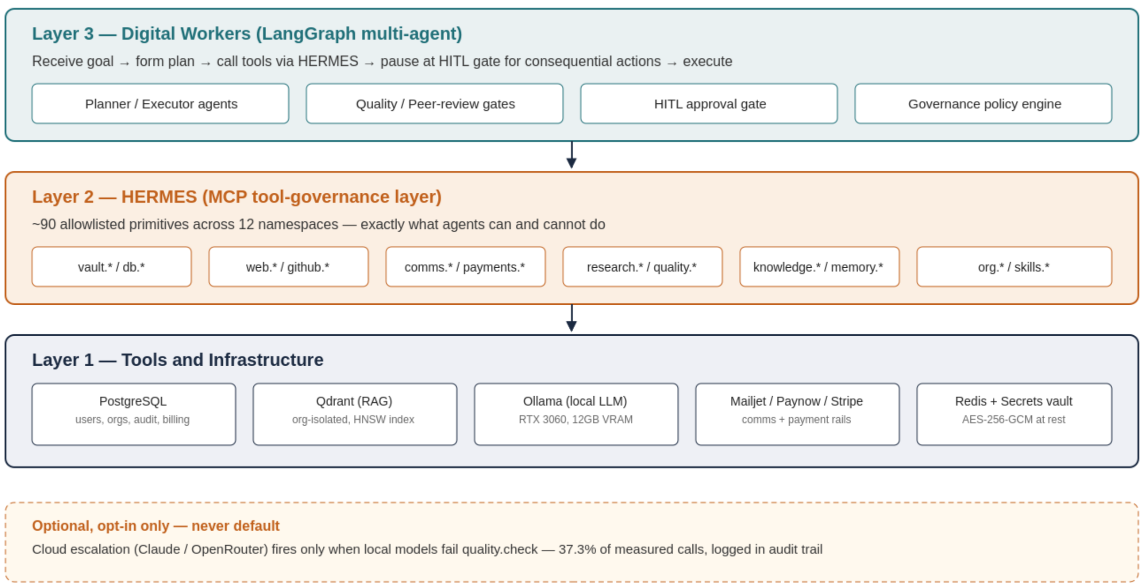
5.4 Scaling Pathway

The formal pilot (Weeks 7–12 post-award) targets 10–20 organisations in Mutare and Harare across SMEs, NGOs, government-adjacent institutions, and one health facility, measured on task completion rate, time saved, cost per task, and at least one non-technical operator case study. SADC regional expansion follows the same architecture with country-specific language/regulatory packs — configuration, not a rebuild. For sovereign government deployment, on-premise with ZCHPC GPU infrastructure avoids cloud dependency; a source-code escrow arrangement is a reasonable structural safeguard against vendor-discontinuation risk and is disclosed here as a commitment to negotiate, not a completed mitigation.

Appendix A — System Architecture Diagram

Not counted towards the 10-page limit. Referenced from Section 2.1.

NeuroWorks — System Architecture



Appendix A — referenced from Section 2.1, System Architecture

Appendix B — Asset and Licence Register (extract)

Not counted towards the 10-page limit. Referenced from Section 4.5 / 5.3. Full register maintained at docs-vault-ready/06-grand-challenge-2026/ in the repository.

Asset	Type	Licence / terms
Ollama runtime + Llama 3.2 / Qwen 2.5 / Llama 3.3 weights	Local inference	Open-weight, per-model licence (Meta Llama Community Licence; Apache 2.0 for Qwen)
Claude Sonnet 5 / Haiku 4.5 (Anthropic API)	Cloud inference, opt-in	Commercial API terms, billed at cost + 10% platform fee
OpenRouter	Cloud routing fallback	Commercial API terms
CLIP, Whisper	Open-weight multimodal models	MIT
PostgreSQL, Redis, Qdrant	Data infrastructure	Open-source (PostgreSQL Licence, BSD, Apache 2.0)
Stripe, Mailjet, Paynow Zimbabwe SDKs	Payment / comms rails	Commercial API terms
Awesome curated-software dataset	IntelliNexus-ingested dataset	CC0
NeuroWorks application code	Proprietary	All rights reserved, Aiia Rubiem / RUBIEM Developers
Zimbabwe sector knowledge packs (planned)	Dataset	CC BY 4.0 on release